

Beam Weapons

While the tractor beam is generally associated with *Star Trek*, beam weapons are a common feature of almost all sci-fi movies and shows. From hand-held phaser weapons to giant planet destroying death stars, a beam weapon has boldly ventured into everyone's imagination. In



1890, Nikola Tesla showcased a machine that was pretty much a man-made lightning creator; transmitting energy without wires, as a high-voltage discharge. This kind of a discharge is chaotic though, since it can travel to pretty much any surface that has a lower voltage charge. In 2007 though, a US company called Ionatron announced that it had won an almost \$10 million contract from the United States Army to create a modern version of that Tesla concept. According to Ionatron, its machine is a far more effective weapon that can disable both people and vehicles—available in both lethal and non-lethal versions, much like setting a phaser to stun or kill. Ionatron's improvement on the Tesla concept was centered on the addition of a special laser. This laser acts as a guide to the electrical discharge—serving to both speed up the discharge and to precisely control where the man-made lightning goes.

In some ways, the design echoes a similar creation by the US Air Force. The Air Force gun however, uses a laser to temporarily blind the enemy. The gun, developed by the force's Research Laboratory's Directed Energy Directorate and about the size and weight of an M60 machine gun, is called the Personnel Halting and Stimulation Response, (PHaSR). The PHaSR shoots a low-power beam of laser light to temporarily blind an individual.

Tractor Beam

Made famous by *Star Trek*—tractor beams are electromagnetic rays that hold an object in space and can also be used to pull objects towards the source of the beam. Such technology is already in use but at a microscopic scale. In 2007, MIT researchers demonstrated such a beam to pickup, hold, and move around individual cells and objects on the surface of a microchip. The beam itself consisted was a laser ray which could be used to push and pull, in a manner similar to how one might use a pair of tweezers.

Tricorder

Here's a common childhood indulgence, influenced by the seminal *Star Trek*. While we spent afternoons communicating with our 'away team' firing make-believe lasers, another

Nano-Tex fibre is being used in clothing to deliver fade-, wrinkle- and stain-resistant fabrics that also keep wearers cool or warm to suit the environment

LumiGram is a company making fibre-optic fabrics that give off radiant light. Materials from this fabric are used in table-cloths (from \$571), tops (\$325), cushions (\$408) and bags (\$211)

MIT is researching materials with fantastical properties – flexible yet highly protective and with muscle-enhancing features, light-refraction for near-invisibility and sensors for constant medical feedback. Crisys anyone?

device, perhaps a little less popular than the communicator or the phaser gun, was the tricorder. Dr. McCoy or science office Spock would frequently use this strange black box to perform tasks ranging from health diagnosis to analysis of alien soil and air.

While the tricorder as it was presented in *Star Trek* is still largely fiction, research towards that has already produced a device that looks like an early prototype to the tricorder. Researchers of Purdue University have created a portable system that can be used to analyze chemical components. Called the Mini 10, this prototype is essentially a portable mass spectrometer, roughly the size of a shoebox; weighing in at 10 kilograms. Running on battery power, the unit weighs 30 times less than a conventional mass spectrometer has applications in everything from airport security to medical diagnostics.

The instrument was developed by a team of researchers led by R. Graham Cooks, the Professor of Analytical Chemistry in Purdue's College of Science. The Mini 10 sniffs for compounds that serve as signatures to the presence of substances such as explosives or cocaine. If these residue indicators are found, the Mini 10 performs a more in-depth analysis to verify the presence of the substance being scanned for.



Another construct from the medical labs of sci-fi shows is the wand that seals wounds and heals without touching the surface of the body. Something similar is tested by the engineers at the University of Washington, employing ultrasound. This technique uses several lenses to focus ultrasound rays to a particular spot—similar to focusing sun rays using a magnifying glass. Such a focused beam of sound can create an extremely hot surface area which would heat up the cells forming the surface of a wound, causing them to seal and cauterise.

Railgun

Made famous by the id Software classic *Quake*, the US Navy has revealed that it's working on a similar device. An 8-megajoule railgun was both created and fired by the Navy in 2007. The gun was also showcased at the Naval Surface Warfare Center, Virginia. The 8-megajoule, electricity guzzling monster-gun can propel non-explosive projectiles at incredible speeds. The weapon is

powerful enough to equal the damage inflicted by a Tomahawk cruise missile, and a range between 200 and 250 nautical-miles. The US Navy plans to deploy a 64-megajoule version by 2020.

Body Armour

From the comic book's *Iron Man* to the protagonist of the video-game *Crysis*—wearing space-age armour that looks cool, deflects bullets, and augments strength and speed has been a staple of science fiction diet. Today, research into nanometer manufacturing and fabrication has made some progress to turning that fiction into fact. Today, materials made up of carbon nanotubes are considerably stronger than steel, while yet good conductors of electricity. Such materials, while in the research stage now, could eventually lead to lighter bulletproof clothing and personal armour. In May of 2007, a start-up called Nanocomp, based in New Hampshire, US, developed nanotubes of length greater than usual. Such longer "fibers" allow the carbon tubes to bind together more effectively resulting in a form of unwoven matting. This matting is then treated chemically to cause the tubes to align, imparting the material with further strength in the direction of alignment. The tubes are made through chemical vapour deposition which involves condensing carbon out of a gas and into the matting. Sheets roughly one metre by three were created using this method. The resulting textile is claimed to be seven times stronger than steel of the same weight.

While armour is used for defense what about a body suit to augment strength? The HAL-5 Exoskeleton (HAL for hybrid assistive limb) is being developed by Professor Yoshiyuki Sankai at Tsukuba University and will likely go into commercial production in this year itself! The suit is essentially a motor-driven metal exoskeleton that you strap onto your body to assist and augment the wearer's motions. So, what's so great about the HAL-5 suit? It augments the wearer's strength, allowing the user to double his lifting abilities. The HAL-5 operates by sensing weak electrical impulses from the user's muscles via electrodes on operator's skin. These impulses are then routed to an onboard computer which analyses the signals for muscle of origin and activates corresponding servos in the HAL-5 suit. It thus mimics and magnifies the wearer's motions.

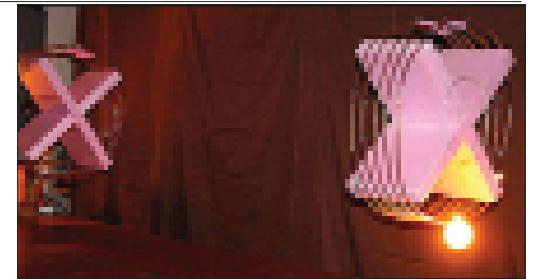
Wireless Electricity

Researches at Massachusetts Institute of Technology recently demonstrated a technology that can wirelessly transfer electricity across a room. The team at MIT showcased the technology which they call WiTricity to power a 60 Watt light bulb from a power source two meters away. The science behind the magic is based on the fact that two objects at the same resonant frequency can effectively exchange energy. The WiTricity setup consisted of two copper coils, oscillating at the same frequency and trading energy across the two meter divide via their electromagnetic field. While one of the coils was attached to a power supply, acting as the transmitter, the other one was connected to the bulb and acted as the receiver. The range for the

Toshiba has been developing robots to aid the elderly—dubbed "life-support partner", these robots will aid elderly and children at home and in public places

A startup company—Emotiv Systems, has a bicycle-helmet-style game controller, dubbed Project Epoc that monitors brain signals and converts thoughts into video-game movements

Japanese telecommunications giant NTT DoCoMo has achieved data rates of 10 megabits per second over skin—soon you will be able to exchange data with just a handshake



WiTricity, however, would probably not extend much beyond a room, a claim we hope the researchers are mistaken in making.

Immortality

The ultimate human dream—to live forever. Immortality could, in theory, be achieved if the cells that comprise our body do not age or destroy themselves. One possible step to this brave new world would be to create a nano-machine which would constantly monitor and regulate cellular behaviour—a guard-dog for each of our cells.

Researchers at Harvard and Princeton University took the first small step towards this goal. They have created implantable devices that can monitor the activities and characteristics of human cells. These machines are constructed of the same material our cells are made up of: DNA, RNA, and proteins. The machines could eventually revolutionise our way of living by directly treating only diseased cells or tissues.

The coolest aspect of this technology is that our body itself can be the producer of these guardians. Harvard's Yaakov Benenson, a Bauer Fellow in the Faculty of Arts and Sciences' Center for Systems Biology says of the research that, "[all that needs to be] provided is a genetic blueprint of the machine and our own biology will do the rest. Your cells will literally build these biocomputers for you."

These nanocomputers will use programmed logarithms to detect anything from the presence of a mutated gene to the activity of genes within the cell. Such machines could, in time, allow doctors to target only cancerous cells, leaving healthy cells completely unaffected.

The Orgasmatron

In Philip K. Dick's highly-influential *Do Androids Dream of Electric Sheep*—the author introduces a device that allows people to literally "dial-in" a mood to change the way they feel. While such a switch would be an extremely welcome addition next to the ceiling fan's knob—the best we have come to achieve is an accident that mimics the Orgasmatron from Woody Allen's movie *Sleeper*.

When anesthesiologist Dr. Stuart Meloy was working on a device for treating chronic pain, he discovered a technique that successfully produced orgasms in women instead. The accidental discovery was made as the doctor powered up an electrode that had been placed into the spine of a woman with chronic back pain.

Based on this discovery, Dr. Meloy hopes to create a device which would allow women to dial-an-orgasm at the touch of a button. The device will have to be surgically implanted at the base of the spine, however. ■

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